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NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



- The area included between the curve $y = x^3$ and $y = 4x^2$ is:
NIMCET 2007
(a) 64 (b) $64/3$
(c) $64/9$ (d) none
- The area included between the curves $y = xe^x$ and $y = xe^{-x}$
NIMCET 2007
(a) 0 (b) -1
(c) 1 (d) none
- The area of the region bounded by the curves $y = |x - 2|$, $x = 1$, $x = 3$ and the x-axis is:
NIMCET 2007
(a) 3 (b) 2
(c) 1 (d) 4
- The area of the plane bounded by the curves $y = \sqrt{x}$, $x \in [0, 1]$, $y = x^2$, $x \in [1, 2]$ and $y = -x^2 + 2x + 4$, $x \in [0, 2]$ is
NIMCET 2008
(a) $\frac{10}{7}$ sq. units (b) $\frac{19}{3}$ sq. units
(c) $\frac{3}{5}$ sq. units (d) $\frac{4}{3}$ sq. units
- A square with side 'a' is revolved about its centre through 45° . What is the area common of both the squares?
NIMCET 2009
(a) $2(\sqrt{2} - 1)a^2$ sq. units (b) $\frac{(\sqrt{2}+1)a^2}{2}$ sq. units
(c) $(\sqrt{3} - 1)a^2$ sq. units (d) $(\sqrt{5} - 1)a^2$ sq. units
- The smaller of the areas bound by $y = 2 - x$ and $x^2 + y^2 = 4$ is
NIMCET 2009
(a) $(\pi - 1)$ sq. units (b) $(\pi - 2)$ sq. units
(c) $(2\pi - 1)$ sq. units (d) $(2\pi - 1)$ sq. units
- The area between the curves $y = 2 - x^2$ and $y = x^2$ is
NIMCET 2010
(a) $\frac{8}{3}$ sq. units (b) $\frac{4}{3}$ sq. units
(c) $\frac{2}{3}$ sq. units (d) $\frac{5}{3}$ sq. units
- The area enclosed within the lines $|x| + |y| = 1$ is
NIMCET 2011
(a) 1 (b) 2 (c) 3 (d) 4
- If the area bounded by $y = x^2$ and $y = x$ is A sq. units, then the area bounded by $y = x^2$ and $y = 1$ is
NIMCET 2011
(a) $(2A + 1)$ sq. units (b) $2A$ sq. units
(c) $(2A + 2)$ sq. units (d) $(A + 2)$ sq. units

- The area enclosed within the curve $|x| + |y| = 1$ (in sq. units) is
NIMCET 2013
(a) $\sqrt{2}$ (b) 1 (c) $\sqrt{3}$ (d) 2
- The area of the region bounded by the lines $y = |x - 1|$ and $y = 3 - |x|$ is
NIMCET 2016
(a) 3 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 2 sq. units
- The area enclosed between the curves $y^2 = x$ and $y = |x|$ is
NIMCET 2018
(a) $2/3$ sq. units (b) 1 sq. units
(c) $1/6$ sq. units (d) $1/3$ sq. units
- Find the area bounded by the line $y = 3 - x$, the parabola $y = x^2 - 9$ and $x \geq -1$, $y \geq 0$.
NIMCET 2020
(a) $7/2$ (b) $11/2$
(c) $9/2$ (d) None of the above
- The area enclosed between the curves $y^2 = x$ and $y = |x|$ is
NIMCET 2020
(a) $2/3$ sq. units (b) 1 sq. units
(c) $1/6$ sq. units (d) $1/3$ sq. units
- The area of the region bounded by the X-axis and the curves defined by $y = \tan x$, $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$ and $y = \cot x$, $\frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$ is
NIMCET 2021
(a) $-\frac{1}{2} \log 2$ (b) $\frac{1}{2} \log 2$
(c) $\log 2$ (d) None of these
- The area enclosed within the curves $|x| + |y| = 2$ is
NIMCET 2022
(a) 16 sq. units (b) 24 sq. units
(c) 32 sq. units (d) 8 sq. units
- The area enclosed between the curve $y = \sin x$, $y = \cos x$, $0 \leq x \leq \frac{\pi}{2}$ is
NIMCET 2025
(a) $(\sqrt{2} - 1)$ (b) $2(\sqrt{2} + 1)$
(c) $(\sqrt{2} + 1)$ (d) $2(\sqrt{2} - 1)$

ANSWER KEY

1.	b	2.	a	3.	c	4.	b	5.	a
6.	b	7.	a	8.	b	9.	a	10.	d
11.	b	12.	c	13.	d	14.	c	15.	d
16.	d	17.	d						



Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	03	NIMCET 2016	01
NIMCET 2008	01	NIMCET 2017	00
NIMCET 2009	02	NIMCET 2018	01
NIMCET 2010	01	NIMCET 2019	00
NIMCET 2011	02	NIMCET 2020	02
NIMCET 2012	00	NIMCET 2021	01
NIMCET 2013	01	NIMCET 2022	01
NIMCET 2014	00	NIMCET 2023	00
NIMCET 2015	00	NIMCET 2024	00
		NIMCET 2025	00

SOLUTION

1. (b) Area enclosed between $y = x^3$, $y = 4x^2$

$$x^3 = 4x^2$$

$$x^3 - 4x^2 = 0$$

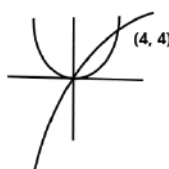
$$x^2(x-4) = 0 \quad x=0, y=0$$

$$x=4, y=4$$

$$\text{Area} \left| \int_0^4 (x^3 - 4x^2) dx \right|$$

$$= \left[\frac{x^4}{4} - \frac{4 \times x^3}{3} \right]_0^4$$

$$= 4^4 \left[\frac{3-4}{12} \right] = \frac{4^4}{3} = \frac{64}{3} \text{ Ans.}$$



2. (a)

$$y = xe^x$$

$$y = xe^{-x}$$

$$xe^x = xe^{-x}$$

$$(e^x)^2 = 1$$

$$e^x = 1$$

$$x = 0, y = 0$$

So there is no Area included between them.

3. (c) $y = |x-2|$, $x = 1$, $x = 3$

Area of ABCDE

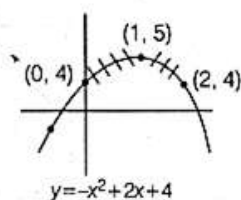
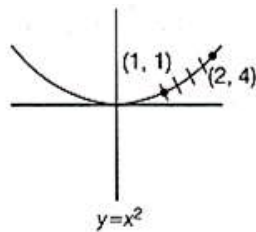
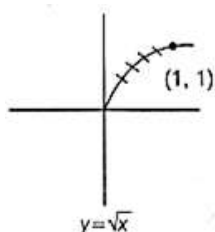
$$= 2 \text{ Area of } \triangle ABC$$

$$= 2 \left(\frac{1}{2} AB \times BC \right)$$

$$= 2 \times \frac{1}{2} \times 1 \times 1$$

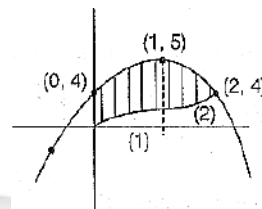
$$= 1$$

4. (b)



$$A_1 = \int_0^1 \sqrt{x} dx = \left(\frac{2x^{3/2}}{3} \right)_0^1 = \frac{2}{3} = \left(\frac{x^3}{3} \right)_1^2 = \frac{7}{3}$$

$$\begin{aligned} A_3 &= \int_0^2 (-x^2 + 2x + 4) dx \\ &= \left[-\frac{x^3}{3} + x^2 + 4x \right]_0^2 \\ &= \frac{-8}{3} + 4 + 8 = \frac{-8}{3} + 12 \\ &= \frac{28}{3} \end{aligned}$$

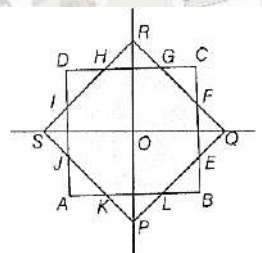


$$\begin{aligned} A_{\text{final}} &= A_3 - (A_1 + A_2) \\ &= \frac{28}{3} - \left(\frac{2}{3} + \frac{7}{3} \right) \end{aligned}$$

$$= \frac{28}{3} - 3$$

$$= \frac{19}{3} \text{ sq. units}$$

5. (a)



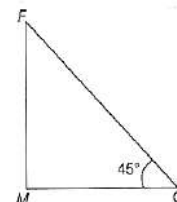
Side length of square is a , hence its diagonal is $\sqrt{2}a$.

Let original square is ABCD and when it rotates by 45° it will be PQRS. Area is required of the shaded part EFGHIJKL (Regular Octagon) = Area of square PQRS - 4 × Area of triangle FDG

So, we have to find area of triangle EQF

$$OQ = \frac{\sqrt{2}a}{2}, \quad OM = a/2$$

$$\text{Hence, } MQ = \frac{a}{\sqrt{2}} - \frac{a}{2}$$



$$MQ = FM = \frac{a}{2\sqrt{2}}(\sqrt{2} - 1)$$

$$FQ^2 = 2 \times \frac{a^2}{8}(\sqrt{2} - 1)^2 = \frac{a^2}{4}(\sqrt{2} - 1)^2$$

$$\Rightarrow FQ = \frac{a}{2}(\sqrt{2} - 1)$$

As, EFQ triangle is a right angle isosceles.

$$\text{Area of } \triangle EFQ = 2 \left[\frac{1}{2} \left[\frac{a}{2}(\sqrt{2} - 1) \right]^2 \right] = \frac{a^2}{4}(3 - 2\sqrt{2})$$

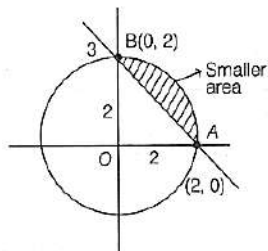
$$\begin{aligned} \text{Area of required octagon} &= a^2 - 4 \left[\frac{a^2}{4}(3 - 2\sqrt{2}) \right] \\ &= 2(\sqrt{2} - 1)a^2 \text{ sq. units.} \end{aligned}$$



6. (b)

$$\begin{aligned} \text{Area of quarter part of circle} &= \frac{\pi r^2}{4} \\ &= \frac{\pi \times 4}{4} = \pi \end{aligned}$$

$$\text{Area of } \triangle OAB = \frac{1}{2} \times 2 \times 2 = 2$$

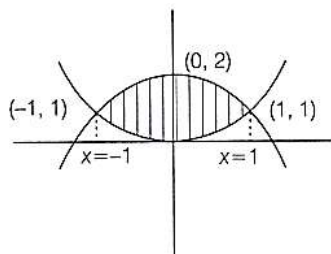


$$\text{Area required} = (\pi - 2) \text{ sq. units}$$

7. (a) $y = 2 - x^2$ and $y = x^3$

$$x^2 = 2 - y$$

$$x^2 = -(y - 2)$$



Point of intersection

$$y = 1, x = \pm 1$$

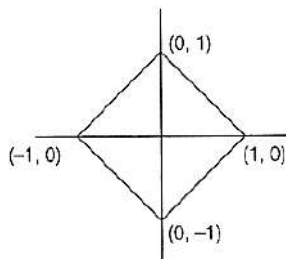
$$A = \int_{-1}^1 ((2 - x^2) - x^2) dx = \int_{-1}^1 (2 - 2x^2) dx$$

$$= 2 \int_{-1}^1 (2 - 2x^2) dx = 2 \left[2x - \frac{2x^3}{3} \right]_0^1$$

$$= 4 \left[1 - \frac{1}{3} \right] = \frac{8}{3}$$

$$A = \frac{8}{3} \text{ sq. units}$$

8. (b)



$x \rightarrow |x| \rightarrow$ symmetric about Y-axis-

$$|x| + y = 1$$

$y \rightarrow |y| =$ symmetric about X-axis-

$$|x| + |y| = 1$$

Formed $\rightarrow$ square

$$\text{Area} = l \times b$$

$$\sqrt{2} \times \sqrt{2} = 2 \text{ sq. units}$$

Short Trick

The area bounded by $|x| + |y| = a$,

$$a > 0$$

$$\text{Area} = (\sqrt{2}a)^2 \text{ sq. unit}$$

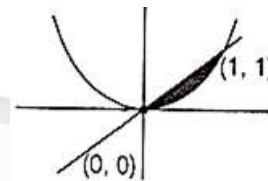
In this question $a = 1$

$$\text{Hence, area} = (\sqrt{2} \times 1)^2 = 2 \text{ sq. units}$$

9. (a)

For $y = x^2$ and $y = x$

Intersection point



$$y = y^2$$

$$y = 0, 1$$

$$x = 0, 1$$

$$A = \int_0^1 (x - x^2) dx$$

$$\text{Area covered by parabola} = \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_0^1$$

$$A_2 = \int_{-1}^1 x^2 dx = \left(\frac{x^3}{3} \right)_{-1}^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \text{ sq unit}$$

$$A_{\text{required}} = \text{Area of rectangle} - \text{Area of covered by parabola}$$

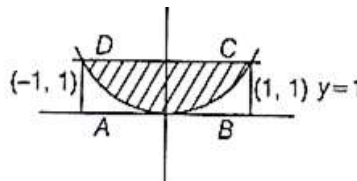
$$A_r = 2 - \frac{2}{3} = \frac{4}{3}$$

$$= 1 + 1/3$$

$$= 1 + 2A \text{ sq.units}$$

for $y = x^2$ and $y = 1$

Intersection point



$$x = \pm 1$$

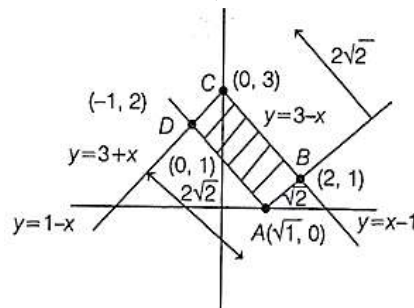
$$\text{Area of rectangle } ABCD = l \times b$$

$$A = 2 \times 1 = 2$$

10. (d) Refer to solution 5.

$$11. (b) y = |x - 1| = \begin{cases} x - 1; & x \geq 1 \\ -x + 1; & x < 1 \end{cases}$$

$$y = 3 - |x| = \begin{cases} 3 - x; & x \geq 0 \\ 3 + x, & x < 0 \end{cases}$$



As AD is parallel to BC and AB is parallel to CD

Here,

$$AB = CD = \sqrt{2}$$

$$BC = AD = 2\sqrt{2}$$

and

$$\text{Area of rectangle } ABCD = l \times b$$

$$= 2\sqrt{2} \times \sqrt{2}$$

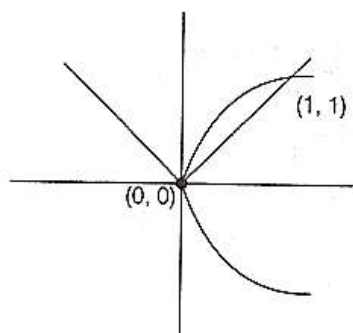
$$= 4 \text{ sq. units}$$





12. (c) Intersection point $\Rightarrow |x|^2 = x$

$$x^2 = x$$



Point of intersections are (0,0) and (1,1)

$$A = \int_0^1 (\sqrt{x} - |x|) dx$$

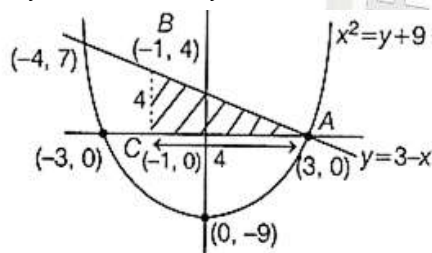
$$A = \int_0^1 (\sqrt{x} - x) dx$$

$$= \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \text{ sq. units}$$

$$A = \frac{1}{6} \text{ sq. units}$$

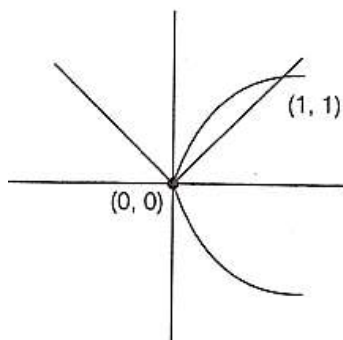
13. (d) $y = x^2 - 9, x \geq -1, y \geq 0$



and $y = 3 - x$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 4 \times 4 = 8 \text{ sq. units}$$

14. (c) Intersection point $\Rightarrow |x|^2 = x, x^2 = x$



Point of intersections are (0,0) and (1,1)

$$A = \int_0^1 (\sqrt{x} - |x|) dx$$

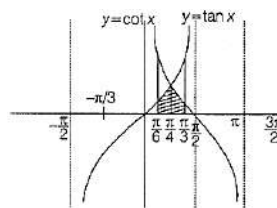
$$A = \int_0^1 (\sqrt{x} - x) dx$$

$$= \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \text{ sq. units}$$

$$A = \frac{1}{6} \text{ sq. units}$$

15. (d)



Area of shaded region

$$A = \int_{\pi/6}^{\pi/4} (\tan x) dx + \int_{\pi/4}^{\pi/3} (\cot x) dx$$

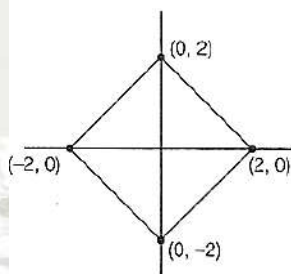
$$= (\log \sec x)_{\pi/6}^{\pi/4} + (\log \sin x)_{\pi/4}^{\pi/3}$$

$$= \left(\log \sqrt{2} - \log \frac{2}{\sqrt{3}} \right) + \left(\log \frac{\sqrt{3}}{2} - \log \frac{1}{\sqrt{2}} \right)$$

$$= \log \frac{3}{2}$$

16. (d) $|x| + |y| = 2 \rightarrow$ Area

$$\text{Area} = l \times b = 2\sqrt{2} \times 2\sqrt{2} = 8 \text{ sq. units}$$



Short Trick:

The area bounded by

$$|x| + |y| = a, a > 0$$

$$\text{Area} = (\sqrt{2}a)^2 \text{ sq. unit}$$

In this question $a = 2$

$$\text{Hence, area} = (\sqrt{2} \times 2)^2 = 2 \times 4$$

$$= 8 \text{ sq. units}$$

17. (d)

Given: Area between $y = \sin x$ and

$$y = \cos x \text{ on } 0 \leq x \leq \frac{\pi}{2}$$

They intersect at $x = \frac{\pi}{4}$.

Area:

$$A = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

First integral:

$$\int_0^{\pi/4} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\pi/4}$$

$$= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) = \sqrt{2} - 1$$

Second integral:

$$\int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx = \sqrt{2} - 1$$

Total area:

$$A = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2(\sqrt{2} - 1)$$

Answer: (d)